SECTIONS OVER A QUASI-COMPACT OBJECT COMMUTE WITH COPRODUCTS AND DIRECT SUMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two external items (the directed-colimits lemma and the description of filtered colimits of abelian sheaves), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Sections over a quasi-compact object commute with filtered colimits with injective transition maps. Stacks Project, Sites, Lemma sites-lemma-directed-colimits-sections (tag 00YQ). Supplies the commutation result to which both statements are reduced.

Source reference. Original reference 0738: sources/stacks-chow/tags.txt, source lines 9052–9052. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 03CO: sources/stacks-chow/tags.txt, source lines 4391–4391. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let W be a quasi-compact object of $\mathcal{C}$.*

(1) *The functor $Sh(\mathcal{C}) \to Sets$, $\mathcal{F} \mapsto \mathcal{F}(W)$ commutes with coproducts.*
(2) *Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. The functor $Mod(\mathcal{O}) \to Ab$, $\mathcal{F} \mapsto \mathcal{F}(W)$ commutes with direct sums.*

Proof. Proof of (1). Taking sections over W commutes with filtered colimits with injective transition maps by Sites, Lemma 0738. If $\mathcal{F}_i$ is a family of sheaves of sets indexed by a set I. Then $\coprod \mathcal{F}_i$ is the filtered colimit over the partially ordered set of finite subsets $E \subset I$ of the coproducts $\mathcal{F}_E = \coprod_{i \in E} \mathcal{F}_i$. Since the transition maps are injective we conclude.

Proof of (2). Let $\mathcal{F}_i$ be a family of sheaves of $\mathcal{O}$-modules indexed by a set I. Then $\bigoplus \mathcal{F}_i$ is the filtered colimit over the partially ordered set of finite subsets $E \subset I$ of the direct sums $\mathcal{F}_E = \bigoplus_{i \in E} \mathcal{F}_i$. A filtered colimit of abelian sheaves can be computed in the category of sheaves of sets. Moreover, for $E \subset E'$ the transition map $\mathcal{F}_E \to \mathcal{F}_{E'}$ is injective (as sheafification is exact and the injectivity is clear on underlying presheaves). Hence it suffices to show the result for a finite index set by Sites, Lemma 0738. The finite case is dealt with in Lemma 03CO (it holds over any object of $\mathcal{C}$). □

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